DGCA EXAMINATION PRACTISE PAPER

1. A Rhumb Line cuts all meridians at the same angle. This gives:-

a) The shortest distance between two points.
b) A line which could never be a great circle track
c) A line of constant direction

2. What is the dip angle at the South Magnetic Pole?

a) 0°
b) 90°
c) 180°
d) 64°

3. If variation is West; then:

a) True North is West of Magnetic North
b) Compass North is West of Magnetic North
c) True North is East of Magnetic North
d) Magnetic North is West of Compass North

4. An airfield has runways 18/36 and 07/25, all of equal length. The TAF W/V is 310/25. Variation is 20°E. Which will be the best R/W for take-off and landing?

a) 18
b) 36
c) 07
d) 25

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5. On the approach to London Heathrow runway 27, glide slope 3°, you reduce speed from 150 kts to 120 kts. What change should you make to your ROD to maintain glideslope?

6. A flight is to be made along the parallel of latitude from A at 48°00'N 04°00'W to B at 48°00'N 02°27'E. Calculate the distance.

7. An aircraft flies for 1000nm along a rhumb line track of 090°(T) from C at 36°00'N 174°45'E to D. What is the longitude of D?

8. Your position is 5833N 17400W. You fly exactly 6 nm eastwards. What is your new position?

a) 5833N 17411.5W
b) 5833N 17355W
c) 5833N 17340W
d) 5833N 17348.5W

9. On a chart, 49 nautical miles is represented by 7.0 centimetres. What is the scale?

a) 1 / 700,000
b) 1 / 2,015,396
c) 1 / 1,296,400
d) 1 / 1,156,600

10. On a particular Direct Mercator wall chart, the 180°W to 180°E parallel of latitude at 53°N is 133 cm long. What is the scale of the chart at 30°S?

a) 1: 3,000,000
b) 1: 18,000,000
c) 1: 21,000,000
d) 1: 27,000,000

11. A blocked pitot probe will affect?
a. ASI.
b. VSI.
c. Altimeter.
d. All of the above.

12. The hysteretic error of an altimeter varies substantially with the?
a. Time passed at a given altitude.
b. Mach number of the aircraft.
c. Aircraft altitude.
d. Static temperature.

13. If ONH is 999 hPa, what is the pressure altitude at an elevation of 25000 ft?
a. 25100ft.
b. 25200 ft.
c. 25300 ft
d. 25400 ft.

14. At 60S on a Mercator chart, chart convergence is:

a) greater than Earth convergency
b) "correct"
c) less than Earth convergency
d) equal to ch long x 0.866

15. On a Lambert's chart the constant of the cone is 0.80. A is at 53N 04W. You plan to fly to B. The initial Lambert's chart straight-line track is 070°(T) and the Rhumb Line track from A to B is 082°(T). What is the longitude of B?

a) 26E
b) 34W
c) 11E
d) 15E

16. SAT = +35°C. Pressure alt (corrected for barometric error) = 5000 feet. What is true alt?

(2 Marks)

a) 4550 feet
b) 5550 feet
c) 4920 feet
d) 5320 feet

17. What is the approximate compression of the Earth?

a) 3%
b) 0.03%
c) 0.3%
d) 1/3000

18. You plan to fly a track (course) of 348°(T), Drift is 17°port, Variation = 32°W. Deviation is 4°E. What compass heading should you fly?

(2 Marks)

a) 041°
b) 033°
c) 016°
d) 359°

19. At a specific location, the value of magnetic variation

(1 Mark)

a) depends on the value of magnetic heading
b) depends on the value of true heading
c) varies slowly over time
d) depends on the type of compass installed

20. Given:

True Track	=	352°(T)
Variation	=	11°W
Deviation	=	- 5°
Drift	=	10°R

What is Heading °(C)?

a) 078°(C)
b) 346°(C)
c) 358°(C)
d) 025°(C)

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21. You are flying at FL330 at Mach No 0.9M. Ambient temperature is ISA $+15^0$. What is your TAS?

(1 Mark)

a) 600 knots
b) 595 knots
c) 540 knots
d) 505 knots

22. You plan to land on R/W 14. The met forecast wind velocity is 110/30. Variation is 30^0W. What crosswind do you expect?

(1 Mark)

a) 15 kts
b) 0 kts
c) 26 kts
d) 30 kts

23. An aircraft is maintaining a 5.2% gradient on a flat terrain. Its height at 7 nm from the runway is approximately?

(2 Marks)

a) 3640 feet
b) 1890 feet
c) 2210 feet
d) 680 feet

24. Course 040^0(T), TAS 120 kn, Wind speed = 30 knots. From which direction will the wind give the greatest drift?

(1 Mark)

a) 215^0(T)
b) 230^0(T)
c) 235^0(T)
d) 240^0(T)

25. An aircraft is climbing at a constant CAS in ISA conditions. What will be the effect on TAS and Mach No?

(1 Mark)

a) TAS increases and Mach No decreases
b) Both increase
c) Both decrease
d) TAS decreases and Mach No increases

26. Convert 70 metres/sec into knots.

(1 Mark)

a) 136 knots
b) 36 knots
c) 146 knots
d) 54 knots

Q.27 What is the ration between the litre and the US-GAL? 1 US-GAL equals?
a. 3.5 litres
b. 4.3 litres
c. 3.78 litres
d. 3.65 litres

Q.28 Given: true track is 348°, drift 17° left, variation 32° W, deviation 4° E. What is the compass heading?
a. 041°
b. 141°
c. 044°
d. 033°

Q.29 An aircraft travels 100 statute miles in 20 min, how long does it take to travel 215 nm?
a. 50 min
b. 46 min
c. 48 min
d. 51 min

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30. Airfield elevation is 1000 feet. The QNH is 988. Use 27 feet per hectopascal. What is pressure altitude?

(1 Mark)

a) 675 feet
b) 325 feet
c) 1675 feet
d) 825 feet

31. A non-perspective chart:

(1 Mark)

a) is produced directly from a light projection of a Reduced Earth
b) cannot be used for navigation
c) is produced by mathematically adjusting a light projection of the Reduced Earth
d) is used for a Polar Stereographic projection

32. The maximum difference between Mean Time and Apparent Time is:

a) 21 minutes
b) 16 minutes
c) 30 minutes
d) there is no difference

33. What is the length of a Sidereal Year?

a) 365 days
b) 366 days
c) 365 days 6 hrs
d) 365 days 5 hrs 48.75 minutes

34. Given: TAS 140 kt, HDG 005°(T), W/V 265/25, calculate the drift and groundspeed.

a) 11R - 140 kt
b) 10R - 146 kt
c) 9R - 140 kt
d) 11R - 142 kt

(2 marks)

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35. Given: FL 120, OAT is ISA, CAS = 200 knots, Track = 222°(M), Heading = 215°(M), Variation = 15°W.

If the time to fly 105 nm is 21 minutes, what is the W/V?

a) 040T / 105 kt
d) 055T / 105 kt
c) 050T / 70 kt
d) 065T / 70 kt (2 marks)

36. Fuel flow per HR is 22 US-GAL, total fuel on board is 83 IMP GAL. What is the endurance ?

a. 4 hr 32 min
b. 3 hr 46 min
c. 4 hr 46 min
d. 3 hr 32 min

37. Given: True track 180° Drift 8°R Compass heading 195° Deviation -2° Calculate the variation?

a. 23°W
b. 20°W
c. 21°W
d. 21°W.

39. Given: true course A to B = 250° Distance A to B= 315 NM TAS=450 kt. W/V=200°/60kt. ETD A= 0650 UTC. What is the ETA at B?

a. 0726 UTC.
b. 0659 UTC.
c. 0732 UTC.
d. 0736 UTC.

40. Given: Runway direction 083° (M), Surface W/V 035/35kt. Calculate the effective headwind component?

a. 24 kt.
b. 20 kt.
c. 22 kt.
d. 26 kt

41. On a Lambert conformal conic, the distance between parallels of latitude spaced the same number of degrees apart?

a. Increases between and reduce outside of the standard parallels.
b. Reduces between, and expands outside, the standard parallels.
c. Is constant between the standard parallels.
d. Is constant outside of the standard parallels.

42. The ‘departure’ between positions 60°N 60°E and 60° N ‘x’ is 1800 NM East. What is the longitude of ‘x’?
a. 145° E.
b. 145° W.
c. 120° E.
d. 120° W.

43. A straight line on a Lambert Conformal Projection chart for normal flight planning purpose?
a. Is a Rhumb line.
b. Can only be a parallel of latitude.
c. Is approximately a Great Circle.
d. Is a Loxodromic line.

44. Isogonals converge at the?
a. Magnetic equator.
b. North magnetic pole only.
c. North and South magnetic poles only.
d. North and South geographic and magnetic poles.

45. What is the angle between True North and Magnetic North?
a. Drift.
b. Variation.
c. Dip.
d. Deviation.

46. **revenue flight is to be made by a jet transport. The following are the aeroplane's structural limits:-Maximum Ramp Mass: 69 900 kg-Maximum Take Off Mass: 69 300 kg-Maximum Landing Mass: 58 900 kg-Maximum Zero Fuel Mass: 52 740 Kg Take Off and Landing mass are not performance limited .Dry Operating Mass: 34 930 kg Trip Fuel: 11 500 kg Taxi Fuel: 250 kg Contingency & final reserve fuel: 1 450 Kg Alternate Fuel: 1 350 kg The maximum traffic load that can be carried is:**

a) 21 170 kg
b) 17 810 kg
c) 21 070 kg
d) 20 420 kg

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47. **With respect to aeroplane loading in the planning phase, which of the following statements is always correct ?**

a. MTOM = ZFM + maximum possible fuel mass
b. MZFM = Traffic load + DOM
c. LM = TOM - Trip Fuel
d. Reserve Fuel = TOM - Trip Fuel

48. 730 FT/MIN equals?

a. 5.2 m/sec.
b. 1.6 m/sec.
c. 2.2 m/sec.
d. 3.7 m/sec.

49. If there is a 15 knot increase in headwind by what amount must the rate of descent be changed in order to maintain a 3° glideslope?

a. It must be decreased by 79 ft/ min.
b. It must be increased by 79 ft/ min.
c. It must be increased by 35 ft/ min.
d. It must be decreased by 35 ft/ min.

50. At 0422 an aircraft at FL370, GS 320kt, is on the direct track to VOR 'X' 185 NM distant. The aircraft is required to cross VOR 'X' at FL80. For a mean rate of descent of 1800 FT/MIN at a mean GS of 232 kt, the latest time at which to commence descent is?

a. 0451.
b. 0454.
c. 0445.
d. 0448.

51. An aircraft was over 'Q' at 1320 hours flying direct to 'R'. Given : Distance 'Q' to 'R' 3016 NM, True airspeed 480 kt, Mean wind component 'out' -90 kt, Mean wind component 'back' +75 kt, Safe endurance 10:00 HR . The distance from 'Q' to the Point of Safe Return (PSR) 'R' is?

a. 1510 NM.
b. 2290 NM.
c. 2370 NM.
d. 1310 NM.

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52. How far can the aeroplane fly out from its base and return in one hour, when flying at TAS 260 kt on a track of 090°, if the W/V is 045°/50kt?
a. 158 nm.
b. 128 nm.
c. 188 nm.
d. 175 nm.

53. What is the chart distance between longitudes 179° E and 175°W on a direct Mercator chart with a scale of 1 : 5 000 000 at the equator?
a. 133 mm.
b. 100 mm.
c. 113 cm.
d. 125 mm.

54. What is the meaning of the term "standard time"?
a. It is the time set by the legal authorities for a country or part of country.
b. It is just another term for UTC.
c. It is time related to the Prime meridian.
d. It is constant at all points on the earth.

55. What is the local mean time, position 65°25'N 123°45'W at 2200 UTC?
a. 0615.
b. 1815.
c. 1345.
d. 1415.

56. The main reason that day and night, throughout the year, have different duration, is due to the?
a. The changing distance between the Sun and the Earth.
b. The changing speed of rotation of the earth.
c. The changing shape of the orbit to the Earth around the Sun.
d. The inclination of the ecliptic to the Equator.

57. If QNH is 1000 hPa and field elevation is 4500 ft amsl, what is QFE?
a. 850 hPa.
b. 163 hPa.
c. -850 hPa.
d. 900 hPa.

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58. Pressure altitude is?
a. The altitude above sea level.
b. The altimeter indication when QFE is set on the sub-scale.
c. The altimeter indication when QNH is set on the subscale.
d. The altimeter indication when 1013.25 hPa is set on the sub-scale.

59. Which of the following cause air density to decrease?
a. Increasing humidity, increasing altitude, increasing temperature.
b. Increasing humidity, increasing altitude, decreasing temperature.
c. Increasing humidity, decreasing altitude, increasing temperature.
d. Decreasing humidity, increasing altitude, decreasing temperature.

60. If QNH changes from 1013 hPa to 1022 hPa this will?
a. Increase field elevation.
b. Decrease field elevation.
c. Not affect field elevation.
d. Decrease QFE.

61. What is the true altitude of an aircraft flying at 16000 ft indicated altitude with an OAT of-16 degrees C?
a. 13200ft.
b. 14200 ft.
c. 16050ft.
d. 16200ft.

62 From where does the ADC obtain altitude data?
a. Radio Altimeter.
b. OAT sources.
c. Barometric altitude source.
d. Dynamic minus total pressure.

63.When descending through an inversion at constant CAS?
a. TAS increases.
b. Mach number increases.
c. Mach number remains constant.
d. TAS remains constant

64.What is V_{FE}?
a. The maximum speed for extending or retracting the flaps.

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b. The maximum speed for flight with the flaps extended.
c. The maximum speed for the flight envelope.
d. The minimum speed for extending or retracting the flaps.

65. What is V_{LO}?
a. The maximum speed with landing gear out.
b. The maximum speed for retracting or extending the landing gear.
c. The minimum speed for flight with landing gear out.
d. The minimum speed for flight when operating the landing gear.

66 What does the green arc on an ASI indicate?
a. V_{LE} at the lower end and V_{LO} at the upper end.
b. V_{FE} at the lower end and V_{FO} at the upper end.
c. V_{S1} at the lower end and M_{MO} at the upper end.
d. V_{S1} at the lower end and V_{NO} at the upper end.

67.When, in flight, the needle of a needle and ball indicator is on the left and the ball on the right, the aircraft is?
a. Turning left with too much bank.
b. Turning right with not enough bank.
c. Turning right with too much bank.
d. Turning left with not enough bank.

68. Following 180° stabilized turn with a constant attitude and bank, the artificial horizon indicates?
a. Too high pitch-up and too low banking.
b. Too high pitch-up and correct banking.
c. Attitude and banking correct.
d. Too high pitch up and too high banking.

69. When standing at the South Pole in which direction will you be facing?
a. North.
b. South.
c. East.
d. West.

70. What is the approximate circumference of the earth?
a. 5400 nm.
b. 11800 nm.

c. 21600 nm.
d. 43400 nm.

71. An aircraft was over 'A' at 1435 hours flying direct to 'B'. Given : Distance 'A to 'B' 2900 NM, True airspeed 470 kt, Mean wind component 'out' +55 kt, Mean wind component 'back' -75 kt, The ETA for reaching the Point of Equal Time (PET) between 'A' and 'B' is?
a. 1744.
b. 1846.
c. 1721.
d. 1657.

72. What is the meaning of the term "standard time"?
a. It is the time set by the legal authorities for a country or part of country.
b. It is just another term for UTC.
c. It is time related to the Prime meridian.
d. It is constant at all points on the earth.

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74. The main reason that day and night, throughout the year, have different duration, is due to the?
a. The changing distance between the Sun and the Earth.
b. The changing speed of rotation of the earth.
c. The changing shape of the orbit to the Earth around the Sun.
d. The inclination of the ecliptic to the Equator.

75. Convert 95 meters/ second into knots?
a. 135 kts.
b. 155 kts.
c. 185 kts.
d. 166 kts.

76. If there is a 12 knot increase in headwind by what amount must the rate of descent be changed in order to maintain a 3° glideslope?

a. It must be increased by 35 ft/ min.
b. It must be decreased by 35 ft/ min.
c. It must be decreased by 61 ft/ min.
d. It must be increased by 61 ft/ min.

77. What will be the rate of descent when flying down a 7% glide slope, at a groundspeed of 540 knots?
a. 3130 ft/min.
b. 4830 ft/min.
c. 3830 ft/min.
d. 3740 ft/min.

78. An aircraft flying down a 3° ILS glideslope is at 35 nm DME from the threshold. Using the 1 in 60 rule and approximating 1 nm to 6000 ft, calculate the aircraft height above the runway threshold, assuming that the ILS glidepath crosses the threshold at a height of 50 ft?
a. 6450 ft.
b. 7455 ft.
c. 10550 ft.
d. 10015 ft.

79. Given: Position A 45° N, ?° E, Position B 45 ° N, 45 ° 15'E Distance A-B= 350 NM, B is to the East of A, What is the longitude of position A.
a. 37° 00'E.
b. 48° 00'E.
c. 51° 15'E.
d. 8° 15'E.

80. An increase of 0.15 mach results in an increase of 93 Kts TAS. The local speed of sound is?
a. 560 Kts.
b. 685 Kts.
c. 620 Kts.
d. 580 Kts.

81. Given the following: True track: 192° Magnetic variation: 7°E Drift angle: 5° left What is the magnetic heading required to maintain the given track?

a. 190° M
b. 192° M
c. 197° M
d. 188° M

82. Given the following: Magnetic heading : 060° Magnetic Variation:8°W Drift angle: 4°right, what is the true track?
a. 058°
b. 054°
c. 052°
d. 056°

83. Given FL350, Mach0.80, OAT -55°C. Calculate the values for TAS and local speed of sound (LSS)
a. 460 kt, LSS 575 kt.
b. 460 kt, LSS 570 kt
c. 465 Kt, LSS 575 kt.
d. 460 kt, LSS 567 kt.

84. 265 US-GAL equals? (Specific gravity 0.80)
a. 803 kg.
b. 810 kg.
c. 800 kg
d. 813 kg.

85. Given : TAS= 190 kt. True HDG= 085° ,W/V=110°(T)/ 50 kt, Calculate the drift angle and GS
a. 8°R- 146 kt.
b. 8°L- 146 kt.
c. 8°L- 140 kt.
d. 8°R- 140 kt.

86. Given: Runway direction 230° (T), Surface W/V 280° (T)/40kt. Calculated the effective cross-wind component?
a. 31 kt
b. 25 kts.
c. 33 kts.
d. 34 kts.

87. How do rhumb lines, (other than meridians), appear on Polar Stereographic charts?
a. As straight lines.
b. As lines concave to the nearer pole.
c. As lines convex to the nearer pole.
d. As ellipses around the pole.

88. Where is the convergency correct on a Transverse Mercator chart?
a. At the datum meridian and the Equator.
b. Only at the Equator and Poles.
c. Only at the datum meridian.
d. At the Parallel of Origin.

89. In which of the following projections will a plane surface touch the reduced earth at one of the Poles?
a. Lambert's.
b. Direct Mercator.
c. Transverse Mercator.
d. Stereographic.

90. Conversion of fuel volume to mass

a) may be done by using standard fuel density values as specified in the Operations Manual, if the actual fuel density is not known.
b) may be done by using standard fuel density values as specified in JAR - OPS 1.
c) must be done by using actual measured fuel density values.
d) must be done using fuel density values of 0.79 for JP 1 and 0.76 for JP 4 as specified in JAR

91. Standard masses may be used for the computation of mass values for baggage if the aeroplane

a) is carrying 30 or more passengers.
b) has 6 or more seats.
c) has 30 or more seats.
d) has 20 or more seats

92. Position A is 55° N 030° W and B is 54° N 020° W. What is the rhumb line bearing from A to B, if the great circle track from A to B, measured at A, is 100°T?

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a. 110°T.
b. 284°T.
c. 104°T.
d. 090°T.

93. A revenue flight is to be made by a jet transport. The following are the aeroplane's structural limits:-Maximum Ramp Mass: 69 900 kg-Maximum Take Off Mass: 69 300 kg-Maximum Landing Mass: 58 900 kg-Maximum Zero Fuel Mass: 52 740 Kg Take Off and Landing mass are not performance limited .Dry Operating Mass: 34 930 kg Trip Fuel: 11 500 kg Taxi Fuel: 250 kg Contingency & final reserve fuel: 1 450 Kg Alternate Fuel: 1 350 kg The maximum traffic load that can be carried is:

a) 21 170 kg
b) 17 810 kg
c) 21 070 kg
d) 20 420 kg

94. Given: Chart scale is 1: 5 000 000. The chart distance between two points is 10 centimetres. Earth distance is approximately?
a. 230 nm.
b. 243 nm.
c. 255 nm.
d. 270 nm.

95. The total length of the 70° N parallel of latitude on a direct Mercator chart is 120 cm. What is the approximate scale of the chart at latitude 40°S?
a. 1 : 16 050 000.
b. 1 : 2 600 000.
c. 1 : 7 250 000.
d. 1 : 25 500 000.

96. A straight line drawn on a chart measures 4.63 cm and represents 125 NM. The chart scale is?
a. 1 : 5 000 000.
b. 1 : 6 000 000.
c. 1 : 4 500 000.
d. 1 : 6 500 000.

97. Two aircraft are flying eastwards around the earth. Aircraft A is flying along the 50N parallel of latitude and aircraft B is flying along the equator. If aircraft A is flying at a groundspeed of 340 knots, what would be the groundspeed of aircraft B both aircraft fly once round the Earth in the same journey time?

a. 529 knots.
b. 540 knots.
c. 480 knots.
d. 600 knots.

98. The sun rises at 50° N and 025°W on the 25th January at 0654 UTC. On the same day what time does it rise at 50° N and 040°E?

a. 0254 UTC.
b. 0154 UTC.
c. 0234 UTC.
d. 0514 UTC.

99. What is the longitude of a position 15 NM to the east of 58° 42'N 094° 00'W?

a. 093° 51.3'W.
b. 094° 13.0'W.
c. 093° 31.1'W.
d. 122° 31.1'W.

100. Total Air Temp is always... than Static Air Temp and the difference varies with...

A) colder, altitude.
B) warmer, altitude.
C) warmer, TAS.
D) colder, CAS.